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TREND-BASED MOCK PAPER

GATE Life Sciences

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Life Sciences
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Life Sciences

Q1. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q2. [1-NAT] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q4. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q5. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q6. [1-MCQ] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q7. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q8. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = Cx$
- (B) $y = C/x$
- (C) $y = Cx^2$
- (D) $x^2 + y^2 = C$

Q9. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q10. [1-NAT] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-NAT] [1 mark]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q14. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1

- (B) 2
- (C) 3
- (D) 0

Q15. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q17. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q18. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q19. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q20. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q22. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6

- (C) 36
(D) 12

Q23. [2-MSQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
(B) $\|u\| + \|v\|$
(C) $\|u\| / \|v\|$
(D) $\sqrt{\|u\| * \|v\|}$

Q24. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
(B) Cosine terms
(C) Both
(D) Neither

Q25. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
(B) 8
(C) 16
(D) 12

Q26. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
(B) 8
(C) 16
(D) 12

Q27. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q30. [1-MCQ] [1 mark]

Strongest acid:

- (A) HCl
(B) HNO₃
(C) H₂SO₄
(D) HClO₄

Q31. [1-MCQ] [1 mark]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q32. [1-NAT] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

(Numerical Answer Type — Enter the numerical value)

Q33. [1-NAT] [1 mark]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-NAT] [1 mark]

Galvanic cell converts:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-NAT] [2 marks]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q37. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q38. [2-NAT] [2 marks]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q40. [1-MCQ] [1 mark]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic
- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q41. [1-NAT] [1 mark]

Strongest field ligand (spectrochemical series):

(Numerical Answer Type — Enter the numerical value)

Q42. [1-NAT] [1 mark]

[Co(NH₃)₆]³⁺ coordination number:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties
- (D) Color only

Q44. [1-NAT] [1 mark]

Octahedral crystal field splits d-orbitals into:

(Numerical Answer Type — Enter the numerical value)

Q45. [2-MCQ] [2 marks]

[Co(NH₃)₆]³⁺ coordination number:

- (A) 3
- (B) 4
- (C) 6
- (D) 8

Q46. [2-MCQ] [2 marks]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties
- (D) Color only

Q47. [2-MCQ] [2 marks]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic
- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q48. [2-MCQ] [2 marks]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic

- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q49. [1-MCQ] [1 mark]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

- (A) $\sim 110\text{ kJ/mol}$
- (B) $\sim 56\text{ kJ/mol}$
- (C) $\sim 22\text{ kJ/mol}$
- (D) $\sim 8\text{ kJ/mol}$

Q50. [1-NAT] [1 mark]

$dG = -5\text{kJ/mol}$ at 300K . Equilibrium constant:

(Numerical Answer Type — Enter the numerical value)

Q51. [1-MCQ] [1 mark]

First-order: 50% in 10 min. 90% time:

- (A) 10 min
- (B) 33.2 min
- (C) 20 min
- (D) 5 min

Q52. [1-MCQ] [1 mark]

Van der Waals corrects for:

- (A) Only volume
- (B) Only pressure
- (C) Both volume and pressure
- (D) Temperature

Q53. [1-NAT] [1 mark]

pH of 0.01M HCl :

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MCQ] [2 marks]

pH of 0.01M HCl :

- (A) 1
- (B) 2
- (C) 0.01
- (D) 14

Q55. [2-MCQ] [2 marks]

pH of 0.01M HCl :

- (A) 1
- (B) 2
- (C) 0.01
- (D) 14

Q56. [2-NAT] [2 marks]

First-order: 50% in 10 min. 90% time:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

$\Delta G = -5 \text{ kJ/mol}$ at 300K. Equilibrium constant:

- (A) e^2
- (B) e^{-2}
- (C) 2
- (D) 0.5

Q58. [1-MCQ] [1 mark]

Grignard reacts with:

- (A) Alcohol only
- (B) CO_2 , carbonyl compounds
- (C) Acids only
- (D) Alkanes

Q59. [1-MCQ] [1 mark]

Geometrical isomerism in:

- (A) 1-butene
- (B) 2-butene
- (C) 1-butyne
- (D) 2-butyne

Q60. [1-NAT] [1 mark]

Benzene undergoes:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Hofmann elimination gives:

- (A) Most substituted alkene
- (B) Least substituted
- (C) Mixture
- (D) No product

Q62. [2-MCQ] [2 marks]

$\text{S}_\text{N}2$ rate depends on:

- (A) Substrate only
- (B) Substrate + nucleophile
- (C) Nucleophile only
- (D) Solvent

Q63. [2-NAT] [2 marks]

Benzene undergoes:

(Numerical Answer Type — Enter the numerical value)

Q64. [2-MCQ] [2 marks]

Benzene undergoes:

- (A) Only addition
- (B) Only substitution
- (C) Both
- (D) Elimination

Q65. [2-NAT] [2 marks]

Geometrical isomerism in:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	D	1
2	1MCQ	A	1
3	1MCQ	D	1
4	1NAT	B	1
5	1MCQ	B	1
6	1MCQ	A	1
7	1MCQ	B	1
8	1NAT	C	1
9	1NAT	A	1
10	1NAT	C	1
11	1MCQ	B	1
12	1NAT	A	1
13	1NAT	A	1

14	1MCQ	B	1
15	1MCQ	B	1
16	1MCQ	C	1
17	1NAT	A	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1NAT	B	1
21	1MCQ	B	1
22	1MCQ	B	1
23	1NAT	B	1
24	1MCQ	B	1
25	1NAT	B	1
26	1MCQ	A	1
27	1NAT	C	1
28	1MCQ	B	1

29	1NAT	A	1
30	1NAT	A	1
31	2NAT	A	2
32	2MCQ	B	2
33	2MCQ	B	2
34	2MCQ	B	2
35	2NAT	B	2
36	2MCQ	A	2
37	2MCQ	C	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2NAT	C	2
41	2MCQ	C	2
42	2MCQ	A	2
43	2MSQ	A	2

44	2MCQ	A	2
45	2MCQ	C	2
46	2MCQ	C	2
47	2NAT	B	2
48	2MCQ	B	2
49	2MCQ	A	2
50	2MCQ	B	2
51	2MCQ	B	2
52	2NAT	D	2
53	2MCQ	B	2
54	2NAT	B	2
55	2MCQ	C	2
56	2MCQ	B	2
57	2NAT	B	2
58	2NAT	B	2

59	2MCQ	B	2
60	2MCQ	A	2
61	2NAT	A	2
62	2NAT	D	2
63	2NAT	B	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q2.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q3.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q4.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q5.

Answer: B

Solution:

Grignard: reacts with CO₂ (carboxylic acid), carbonyls, epoxides. Reacts with protic solvents.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q6.

Answer: A

Solution:

Low-spin d₆: t_{2g}^6 , all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q7.

Answer: B

Solution:

$2 = \exp(E_a/R \cdot (1/300 - 1/310))$. $E_a = R \ln(2)/(1/300 - 1/310) = 53.6$ kJ/mol. Closest: 56.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q8.

Answer: C

Solution:

CN⁻ (strongest) > NH₃ > en > NO₂⁻ > F⁻ > Cl⁻ > Br⁻ > I⁻ (weakest).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q9.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q10.

Answer: C

Solution:

6 NH₃ ligands: CN = 6. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: difficult

Q11.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q12.

Answer: A

Solution:

$$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q13.

Answer: A

Solution:

$$dG = -RT \ln(K) \Rightarrow K = \exp(-dG/RT) = \exp(5000/(8.314 \cdot 300)) = e^2.$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q14.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: B

Solution:

$$t_{90} = t_{50} \cdot \log(10)/\log(2) = 10 \cdot 3.32 = 33.2 \text{ min.}$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q16.

Answer: C

Solution:

$(P + a/V^2)(V - b) = RT$. Corrects for intermolecular forces (a) and finite volume (b).

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q17.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x \cdot v$ and $y(1) = 0$ gives $y = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

$$L\{t \cdot f(t)\} = -d/ds(F(s)). \quad F(s) = 1/(s+2). \quad -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: difficult

Q21.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q23.

Answer: B

Solution:

$[H^+] = 10^{-2}$. $pH = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q24.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q26.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q29.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: $d_{x^2-y^2}, d_{z^2}$).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q31.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: B

Solution:

SN2: bimolecular. Rate = k[substrate][nucleophile]. Both affect rate.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q34.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q36.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1, 0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1, 0]^T = [1, 0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \text{ Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

$$\mathcal{L}\{e^{-at}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q46.

Answer: C

Solution:

6 NH_3 ligands: CN = 6. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q47.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q48.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q49.

Answer: A

Solution:

Low-spin d_6 : t_{2g}^6 , all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q50.

Answer: B

Solution:

$[H^+] = 10^{-2}$. $pH = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q51.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q52.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{total}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q53.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q54.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q55.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

$[H^+] = 10^{-2}$. $pH = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q57.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q58.

Answer: B

Solution:

$t_{90} = t_{50} * \log(10)/\log(2) = 10 * 3.32 = 33.2$ min.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q59.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q60.

Answer: A

Solution:

Low-spin d_6 : t_{2g}^6 , all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q61.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{total}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q63.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

$dG = -RT \ln(K) \Rightarrow K = \exp(-dG/RT) = \exp(5000/(8.314*300)) = e^2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q65.

Answer: A

Solution:

$$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$$

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 29 questions, 47 marks
Chemistry: 10 questions, 15 marks
Inorganic Chemistry: 9 questions, 13 marks
Physical Chemistry: 9 questions, 13 marks
Organic Chemistry: 8 questions, 12 marks

Difficulty Distribution:

Easy: 19 | Moderate: 39 | Difficult: 7

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.